

h: hydrophobic class p: polar class H/P alphabet (hp_lehninger2) — 20 amino acids (protein20) k that kmerseek tests for hp_lehninger2 (17–30)

a How kmerseek works: CED-9 against BCL2, hp_lehninger2, k = 17 (kmerseek pair, kmerseek main 00395d7)

1 Encode in two letters

h = AFGILMPVWY
p = CDEHKNRST

CED-9 (280 aa)

MTRCTADNSLTNPA...

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

h p p p p h p p p h p p p h h

2 Find shared 17-letter words

CED-9 BCL2

163 139 p p h h p p h h p h h h h h p h h h

164 140 p h h p p h h p h h h h h p h h h h

165 141 h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

3 Join into one region

CED-9 163 QCPMSYGRLLIGLISFGGFV 181

|| || 5 of 19 identical

BCL2 139 RDGVNWGRIVAFEFEGGVM 157

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

h h p h h p h h h h h h p h h h h h

4 Report both coordinates

UniProt BH1 motif

region kmerseek reports

CED-9 BH1 1 280

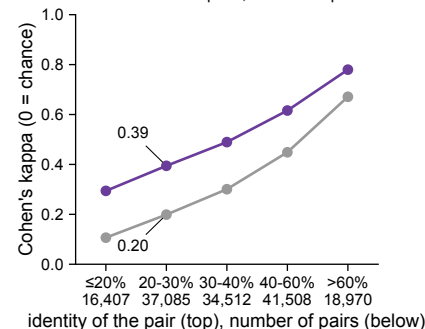
163–181

BCL2 BH1 1 239

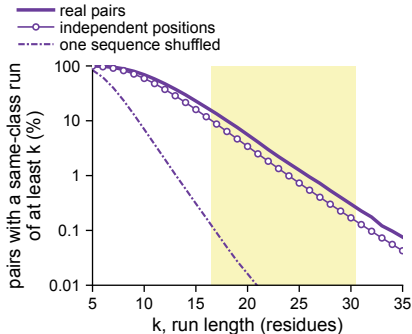
139–157

b Class kept above chance, all pairs

Pfam-A 38.2 seed pairs, mean over pairs



c Same-class runs, pairs at 20-30% identity



d Seed length: homologs vs chance hits

exact hp_lehninger2 seeds; top: Pfam pairs at 20-30% identity; bottom: CED-9 against 19,732 human proteins

